REVIEW NOTES · MANUSCRIPT VERSION 13

Clifford Geometry and Mass Relations

A. Kritov, *Clifford Geometry and Mass Relations: From Charged Leptons to the Proton*, v13. Mathematical check of the manuscript, with its lepton relations placed against Barr–Zee and earlier order-α mass ideas.

3 November 2026 · Equation, section and table numbers refer to the manuscript · Inputs as in its Eq. (8.2), CODATA 2022 and PDG 2026

Summary

MATHEMATICS

The derivations in Sections 2–10 and Appendices A–B are internally consistent. The operator identities, flows, channels and colour identities were checked numerically with the matrices of Eq. (B.1), and every number in Tables 1 and 3 reproduces at 40-digit precision. No error changes a stated result. Four points need attention: the proton comparison leaves out the uncertainty carried by α, which is nine times the reference uncertainty used for its z; the $m_p c^2$ row of Table 3 is not an independent test; the uniqueness claim for the return map in Sec. 6.2 needs one more condition; one figure in Sec. 8.2 should read 8.30×10^{-7}.

CONTEXT

The manuscript belongs to the excitation line of Nambu and Barut, where m_μ/m_e is a leading $1/\alpha$ term with corrections of order one in units of m_e. The radiative line of Weinberg, Georgi–Glashow and Barr–Zee reads the same leading relation from the other side, with m_e generated at order α from m_μ. Read that way, the manuscript fixes the coefficients of α and α^2 in m_e/m_μ. The offset 17/14 is smaller than the shift produced by moving α from $q^2 = 0$ to the muon scale, and smaller than the difference between pole and running mass ratios, so the relation holds as stated only for α(0) and pole masses.

FLEXIBILITY

Section 3 keeps the manuscript's integer and rational choices on simple ratios and lets only α and the energy scale vary. Every discrete choice is then fixed by the data: the nearest simple-ratio neighbour of each misses by at least $1.8 \times 10^4\sigma$. On that grid the muon offset 17/14 is the only match, with a chance probability of about 4×10^{-3} at 1σ. The proton relation has more freedom: its proton-only prescriptions are selected to fit, five screening forms fit equally well once the uncertainty of α is included, and the agreement holds for the CODATA 2022 α but fails by 4.8σ with the Cs recoil value. Running the relations through QED and Standard Model evolution finds no scale better than the infrared: the offsets take the manuscript's values only below a few keV, with α at its Thomson value and pole masses.

KOIDE

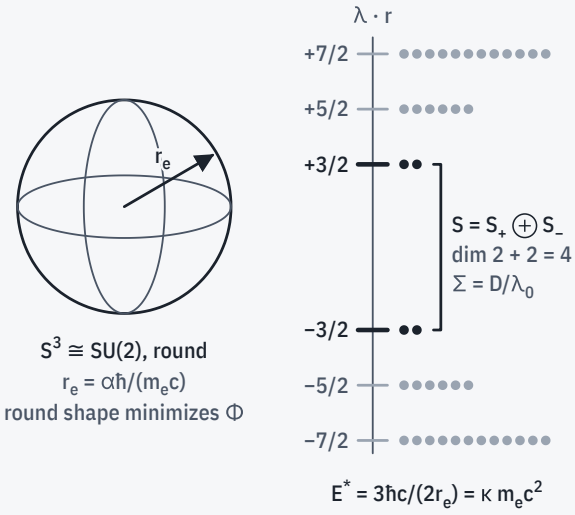
Section 4 follows Koide's ratio through the same parameter space. It is not built into the construction: at the manuscript's point it equals 2/3 to $\mathbf{10^{-6}}$ because the formulas reproduce the measured masses. At fixed weights it drifts with α, crosses 2/3 once at $\boldsymbol{\alpha^{-1} = 137.0455}$, and tends to 2.87% above 2/3 as α → 0. Exact Koide inside the model needs a τ weight of 17.000115, which is not a simple ratio, and running masses move Q further from 2/3. A square-root mass operator with a cyclic Z_3 symmetry gives Koide identically; the manuscript's 1 + 14 + 1 carrier has no such symmetry.

The construction at a glance

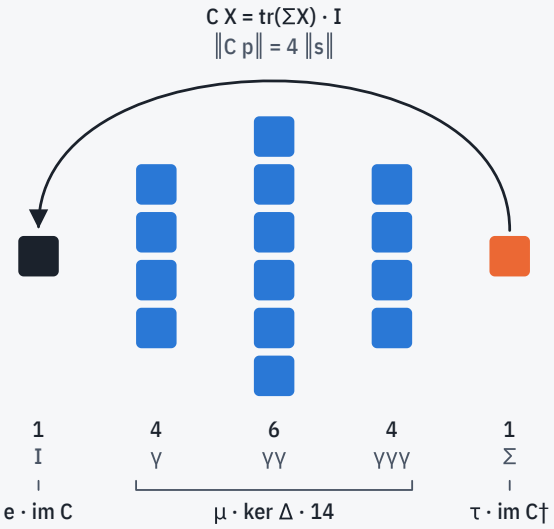
Figures G1–G3 draw what the manuscript builds, where each number in its mass formulas comes from, and how else the carrier could have been split. Equation numbers refer to the manuscript.

FIGURE G1 · FROM THE SPHERE TO THE LEPTON MASSES, EQ. (4.5)

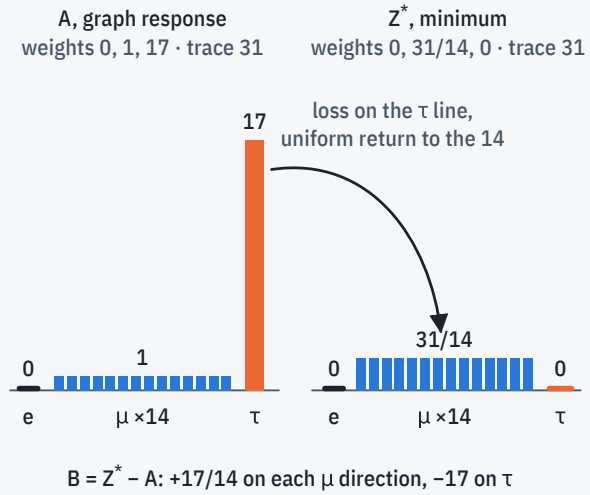
1 · Round S^3 and its lowest Dirac level, Eqs. (2.1)–(2.6)



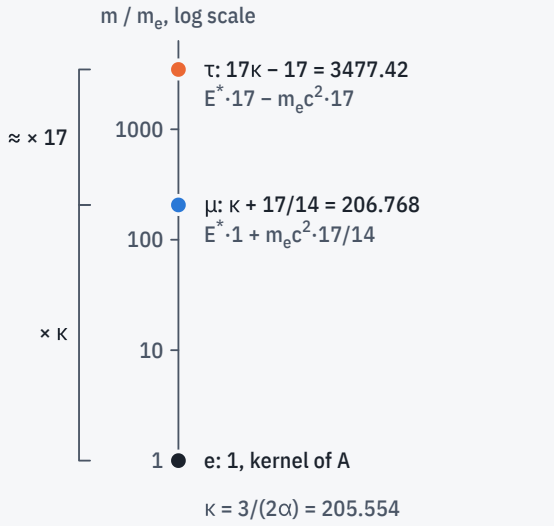
2 · Matrix carrier W = End S and the operator C, Eqs. (3.1)–(3.6)



3 · Response weights and their minimum, Eqs. (5.5), (6.4), (6.13)



4 · Resulting masses, Eqs. (7.3)–(7.4)

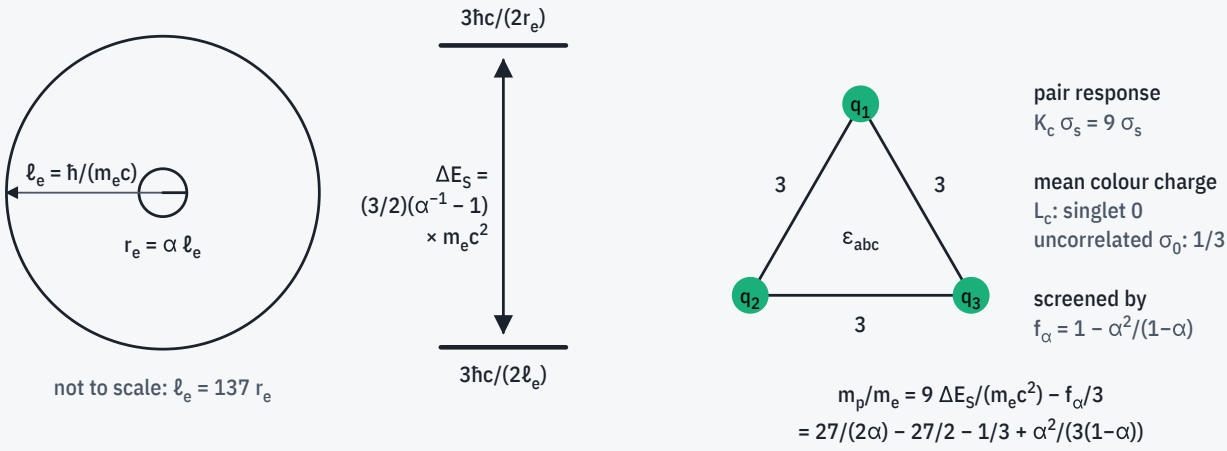


(1) The round S^3 of radius r_e has two Killing spinors of each sign at $\lambda r = \pm 3/2$; they form the carrier S and fix $E_* = \kappa m_e c^2$. (2) The 16 Clifford words of $\mathcal{W} = \mathrm{End}\, S$ split under $\mathcal{C}$ into the scalar line (e), 14 harmonic directions (μ) and the pseudoscalar line (τ); $\mathcal{C}$ maps the pseudoscalar to four times the scalar, which gives the τ weight $1 + 4^2$. (3) The trace-preserving minimum moves the τ weight into the harmonic block. (4) Leading weights times E_* plus the change B in units of $m_e c^2$ give the three masses.

FIGURE G2 · THE PROTON EXTENSION, EQS. (10.1)–(10.12)

Relative energy between two radii, Eqs. (10.6)–(10.7)

Colour singlet and its two responses, Eqs. (10.2)–(10.10)



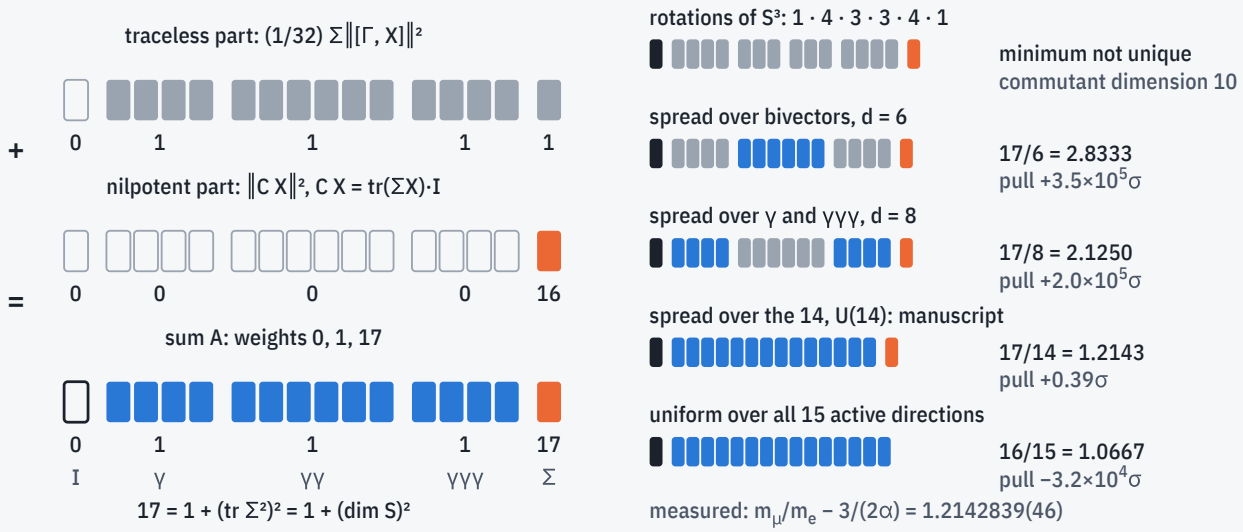
$$m_p/m_e = 9\,\Delta E_S/(m_e c^2) - f_\alpha/3 = 27/(2\alpha) - 27/2 - 1/3 + \alpha^2/(3(1-\alpha))$$

The proton uses the same lowest level, compared between radius r_e and the reduced Compton length $\ell_e = r_e/\alpha$, weighted by the pair response 9 of the colour singlet, minus the screened mean-charge term 1/3. Both prescriptions are specific to the proton; the leptons use the absolute level E_*.

FIGURE G3 · WHERE 0, 1, 17 COME FROM, AND OTHER WAYS TO SPLIT THE 16

The two terms of the response, Eqs. (5.3)–(5.6)

Alternatives to 1 + 14 + 1 and the muon offset 17/d



Left: the traceless part gives 1 to every Clifford word except the identity, and $\|\mathcal{C}X\|^2$ is nonzero only on Σ, where $\mathcal{C}$ multiplies the norm by $\mathrm{tr}\,\Sigma^2 = 4 = \dim S$. Right: rotations of S³ split the 16 words into 1 + 4 + 3 + 3 + 4 + 1 (scalar, vectors, self-dual and anti-self-dual bivectors, pseudovectors, pseudoscalar); with only that symmetry the minimum of Sec. 6 is not unique. The manuscript merges the middle 14 by postulating the full U(14) isotropy (H1). Spreading the τ weight over a block of dimension d gives the muon offset 17/d; spreading the whole trace uniformly over all 15 active directions gives 16/15. Only d = 14 matches.

1 Two readings of $m_\mu/m_e \approx 3/(2\alpha)$

Numerically $m_e/m_\mu = 0.0048363$ and $2\alpha/3 = 0.0048649$, so the leading relation holds to 0.6%. Two lines of work have used it, in opposite directions.

1.1 Excitation line: the heavy leptons as $1/\alpha$ excitations of the electron

1952 **Nambu.** Empirical particle spectrum organized in units of order m_e/α.

1978 **Barut, Phys. Lett. B 73, 310.** A classical relativistic charged particle has anomalous moment $g = 4\alpha/3$; quantizing the dressed particle with a generalized Dirac equation gives a second mass $m_\mu = m_e\,(3/(2\alpha) + 1)$.

1979 **Barut, Phys. Rev. Lett. 42, 1251.** Quantized magnetic self-energy $\frac{3}{2}\alpha^{-1}n^4 m_e$ added level by level: $m_N = m_e\big(1 + \frac{3}{2\alpha}\sum_{n\le N} n^4\big)$. Predicted $m_\tau = 1786.08$ MeV and a fourth lepton at 10293.7 MeV.

1992 **BES, Phys. Rev. Lett. 69, 3021.** Threshold measurement $m_\tau = 1776.9^{+0.4}_{-0.5} \pm 0.2$ MeV, about 9 MeV below Barut's value.

2005 **Mac Gregor, Int. J. Mod. Phys. A 20, 719.** α-quantized masses and lifetimes across leptons and hadrons.

2021 **Kritov, Progress in Physics 17, 158.** α-quantization of particle masses tied to a cosmological entropy bound.

2023 **Scott, Physics Forums.** Observes numerically that $m_\mu/m_e - 3/(2\alpha) \simeq 17/14$.

2026 **Kritov, v13.** manuscript Keeps Barut's leading weights 1 and 17 and replaces his offsets +1, +1 by +17/14 and −17, derived from a Clifford carrier on S^3 with radius r_e.

1.2 Radiative line: the electron mass as an order-α effect of the muon

1972 **Weinberg, Phys. Rev. Lett. 29, 388.** Masses that vanish at zeroth order because of the representation content can be computed as finite higher-order effects. The example generates the electron mass radiatively from the muon; Raby's review notes a flaw in it, later corrected by Georgi and Glashow.

1973 **Georgi and Glashow, Phys. Rev. D 7, 2457.** Only m_μ appears as a counterterm; the electron is massless at zeroth order and acquires a finite mass of order αm_μ, whose value depends on the masses of unobserved vector bosons.

1978 **Barr and Zee, Phys. Rev. D 17, 1854.** A class of models with $m_e/m_\mu = N\,\alpha/(\pi \sin^2\theta_W)$, N a model-dependent number of order one; other corrections are suppressed by making exotic gauge bosons heavy, and the models agree with the neutral-current data of the time. Reproducing the measured ratio needs $N \approx 0.48$ at $\sin^2\theta_W = 0.231$.

1988 **Balakrishna, Phys. Rev. Lett. 60, 1602; Balakrishna, Kagan and Mohapatra, Phys. Lett. B 205, 345.** Generations acquire mass one by one at successive orders of perturbation theory.

1989 **Babu and Ma, Mod. Phys. Lett. A 4, 1975.** Review of six basic radiative mechanisms, three at one loop and three at two loops.

1990 **Babu, Balakrishna and Mohapatra, Phys. Lett. B 237, 221.** Supersymmetric Pati–Salam model: τ massive at one loop, μ and e at two loops.

2008 **Dobrescu and Fox, JHEP 08, 100.** All masses from top loops: τ at one loop, μ at three loops, e at four.

2010 **Graham and Rajendran, Phys. Rev. D 81, 033002.** Domino mechanism: masses of successive generations arise at successive orders.

2021 **Baker, Cox and Volkas, JHEP 05, 174.** Minimal one-loop radiative muon mass models, best probed by $(g-2)_\mu$ and the Higgs–muon coupling.

1.3 The step from μ to τ

The two lines separate at the τ. The measured $m_\mu/m_\tau = 0.0595$ is about 26 times α/π, so in the radiative line the μ–τ step needs a different coupling or loop order than the e–μ step; the models above put the τ at one loop and the μ at two or three. In the excitation line both heavy leptons sit on the same $1/\alpha$ scale and the τ carries an integer weight: Barut's $1+2^4$, the manuscript's $1+4^2$, both 17. There α enters the μ–τ ratio only as a correction. From Eq. (7.4),

$$\begin{aligned} \frac{m_\tau}{m_\mu} &= \frac{17(\kappa-1)}{\kappa+17/14}, \qquad \kappa = \frac{3}{2\alpha}, \\ &= 17\left[1-\tfrac{31}{21}\alpha+O(\alpha^2)\right] = 16.81795, \end{aligned}$$

against 16.81769(85) from the PDG 2026 τ mass and the CODATA muon mass.

Quantity	Barut 1979	Manuscript, Eq. (7.4)	Measured
m_μ/m_e	206.5540	206.7682845	206.7682827(46)
Offset beyond κ	+1	+17/14 = 1.2142857	1.2142839(46)
m_τ/m_e	3495.418	3477.418	3477.37(18)
Offset beyond 17κ	+1	−17	−17.05(18)
$m_\tau c^2$ (MeV)	1786.15	1776.957	1776.93(9)

Barut's values recomputed with CODATA 2022 α and m_e; he quoted 1786.08 MeV with 1979 inputs. Measured τ entries use PDG 2026; with the CODATA 2022 τ mass the offset is −17.19(24).

2 The manuscript read from the radiative side

2.1 Coefficients of α and α^2

Inverting the muon relation of Eq. (8.1) gives the form a Barr–Zee-type calculation would produce:

$$\frac{m_e}{m_\mu} = \frac{2\alpha/3}{1+\frac{17}{21}\alpha} = \frac{2}{3}\,\alpha - \frac{34}{63}\,\alpha^2 + O(\alpha^3).$$

Loop-generated masses usually carry α/π or $\alpha/4\pi$ times logarithms of heavy masses. In the manuscript the leading coefficient 2/3 comes from the lowest Dirac eigenvalue $3/(2r)$ at $r = r_e$ (Eq. 2.6), with no loop factor. In Barr–Zee form the same leading term requires $N = \frac{2\pi}{3}\sin^2\theta_W \approx 0.48$. There N depends on the

model; in the manuscript the corresponding number is fixed by the sphere and by postulates H1–H2, and the second coefficient −34/63 is fixed by the trace redistribution of Sec. 6.

2.2 Which α and which masses

A derivation that produces m_e/m_μ at order α has to state which α and which masses it predicts. At the size of the offset 17/14 the choice changes the answer. Leading-order estimates:

Change of convention	Shift in the ratio	Offset in the manuscript
α(0) → α(m_μ) in $3/(2\alpha)$; one-loop electron vacuum polarization, $\Delta\alpha = 6.97 \times 10^{-3}$	−1.43 in m_μ/m_e	+1.214
Pole ratio → $\overline{\text{MS}}$ ratio at a common scale; one-loop QED factor $1 + \frac{3\alpha}{2\pi}\ln(m_\mu/m_e) = 1.0186$	+3.84 in m_μ/m_e	+1.214
Same for m_τ/m_e; factor 1.0284	+98.8 in m_τ/m_e	−17

The one-loop mass anomalous dimension is the same for e, μ and τ, so the running ratio does not change between common scales; the shift comes from the pole-to-running conversion at different masses.

Each shift is larger than the offset it is compared with. The manuscript's use of α(0) and on-shell masses (Sec. 9.2) is therefore part of the content of Eq. (8.1), and its agreement at 9×10^{-9} relative holds in that convention only. Koide's relation has the same issue: it holds for pole masses to 10^{-5} and fails by about 0.2% for running masses at M_Z (Xing and Zhang 2006). Sumino (2009) built a U(3)×SU(2) family gauge model in which the family-gauge radiative correction cancels the QED correction to Koide's formula. Sec. 11.3 of the manuscript lists self-energy corrections as further research. Given the numbers above, an account of why α(0) and pole masses are the right objects is needed before 17/14 can be read as a structural coefficient. Sec. 3.7 follows the full scale dependence.

2.3 What the data fix

- **Muon.** The measured offset is 1.2142839(46); 17/14 lies at +0.39σ. There are 64 reduced fractions p/q in [1, 2) with $q \le 14$, so the chance that one lies within ±1σ of a point taken at random is about $64 \times 9.2 \times 10^{-6} \approx 6 \times 10^{-4}$. Because 17/14 was found numerically first (Scott 2023, acknowledged in the manuscript), this figure measures the coincidence the construction sets out to explain.
- **Tau.** The offset is −17.05(18) with the PDG 2026 mass. It is resolved to ±0.18, so an integer offset falls within 1σ with probability about 0.36. The τ relation tests the weight 17 and the sign of the offset more than its value.
- **Sum rule** (11.4). In $14m_\mu + m_\tau = 93m_e/(2\alpha)$ the muon term carries an uncertainty of $6 \times 10^{-5} m_e$ and the τ term $0.18\,m_e$. At present precision it is the τ test.

3 Flexibility: what can be changed to reach the targets

The relations contain no fitted continuous parameter, but each results from a sequence of choices. This section lists those choices, shows which ones the data force, and counts how many alternative formulas the same choices generate. Every number below comes from an enumeration over the alternatives named in the tables, with targets $m_\mu/m_e = 206.7682827(46)$, $m_\tau/m_e = 3477.37(18)$ (PDG 2026) and $m_p/m_e = 1836.152673426(32)$.

3.1 Lepton choice points

Choice	Manuscript	Alternatives enumerated	What the data do
Unit energy c/α: Dirac level times radius	$c = 3/2$: lowest level on S^3, $r = r_e$	$c = 1/2, 1, 2, 5/2, 3$	Only 3/2 leaves an order-one muon offset (1.214); the next best leaves 67.3
τ weight g_τ: dim S and trace normalization in $\mathcal{C}$	$17 = 1 + 4^2$: dim $S = 4$, unnormalized trace	dim S = 2, 8; trace divided by n or n^2; gives 2, 3, 5, 9, 65	Only 17 leaves an order-one τ offset (−17.05); every other weight misses by more than 1600
Redistribution space d for B	14, harmonic sector	15 (active), 16 (all of $\mathcal{W}$), 3 (profiles)	The muon offset g_τ/d, known to 2×10^{-8} of the ratio, selects 14
Calibration of B	$\eta = 1$ in units of $m_e c^2$	$\eta = 1/2, 2$; scale E_* in place of $m_e c^2$	Muon fixes $\eta = 1$ to $\pm 3.8 \times 10^{-6}$; the τ alone fixes it to ±0.010
Offset rule	$(+g/d, -g)$	Barut $(+\eta, +\eta)$; none; $(+g/d, -g + g/d)$; $(+(g-1)/d, -(g-1))$	Muon selects g/d with $g = 17$; the τ then excludes $(+g/d, -g + g/d)$ at 7σ
Scale of α	α(0)	$\alpha(m_\mu)$, $\alpha(m_\tau)$, one loop	Shifts of −1.43 (μ) and −50 (τ), larger than the offsets
Mass scheme	Pole	$\overline{\text{MS}}$ at a common scale	Shifts of +3.84 (μ) and +98.8 (τ)

The knobs of Eq. (11.1), the radius factor λ, the structural weight g and the calibration η, are set to 1 by H1–H2, and the other choice points are integers or simple ratios: 3/2, 17 and 14 for the leptons, 9, 1 and 1/3 for the proton. The scans below keep them on simple ratios, meaning integers, half-integers and fractions with denominator up to 4, with dimensions as integers. Only α and, in Sec. 3.7, the renormalization scale vary continuously.

3.2 Counting lepton alternatives

- **Muon offset.** Offsets of the form $\eta N/D$, with N, D from the integers the carrier supplies {1, 2, 3, 4, 5, 6, 14, 15, 16, 17, 31} and $\eta \in \{1/2, 1, 2\}$, give 137 distinct values in (0, 3]. Only 17/14 lies within 2σ of the data. A random offset in (0, 3] would be matched within 1σ with probability 4.2×10^{-4}, within 2σ with 8.4×10^{-4}.
- **Rational grid.** With g and η on fractions with denominator ≤ 4 up to 3, and d from 1 to 20, the offsets $\eta(1 + 16g)/d$ take 1443 distinct values in (0, 3]. Only 17/14 lies within 2σ. A random offset would be matched within 1σ with probability 4.4×10^{-3}, ten times the figure for the smaller integer set above, since the probability grows with the grid.
- **Joint family.** 6 unit energies × 3 dimensions of S × 3 trace normalizations × 4 redistribution spaces × 3 calibrations × 5 offset rules give 3240 formula pairs. One, the manuscript's, fits both leptons within 2σ. Over 200 000 random targets (m_μ/m_e uniform in [150, 300], m_τ/m_e in [2000, 5000], same uncertainties) the family matched the muon 5 times and both leptons never.
- **Reading.** Inside this grammar the lepton formulas are rigid: the leading terms are forced at the percent level and the muon offset carries a look-elsewhere probability near 10^{-3}. The grammar was written after the values were known, and its size is the input these numbers depend on most: doubling the set of rules or integers roughly doubles the probabilities.

3.3 Proton choice points

Choice	Manuscript	Alternatives enumerated	What the data do
Colour weight W	9, from $K_c\sigma_s = 9\sigma_s$	7 to 11 in steps of 1/2	$\rho_p/\kappa = 8.93$; only $W = 9$ works with the next two choices
Reference length	Compton ℓ_e: subtract $W \cdot 3/2$	none; Bohr a_0; factors 1/2, 2	Compton removes 13.5 of the 13.833 needed. Used for the proton only: applied to the muon it gives 205.268, 1.5 below the data
Mean-charge term m	1/3, from the normalization $3\sqrt{C_F}$ and $\sigma_0 = I_c/27$	$\pm p/q$ with $p, q \le 9$	The data need −0.333315
Screening form $f(\alpha)$ in $-mf$	$1 - \alpha^2/(1-\alpha)$	$1-\alpha^2$, $1-\alpha^2/(1+\alpha)$, $1/(1+\alpha^2)$, $\sqrt{1-2\alpha^2}$, $1-\alpha^2/2$, $1-2\alpha^2$, 1	The data need 0.3356 ± 0.0054 times α^2. Five forms fit within 1σ once α's uncertainty is included; with the reference uncertainty alone the manuscript's form is the only one
α input	CODATA 2022	Cs 2018, Rb 2020, electron $g-2$ 2023	See 3.4

- **Counting.** W × reference length × m × sign × f gives 39 600 formulas. Five fit within 2σ; all have $W = 9$, the Compton subtraction and −1/3, and differ only in f. A random target in [1830, 1842] would be matched within 1σ with probability 3.3×10^{-5}, or 4.2×10^{-6} with the reference-only uncertainty that Table 3 uses.
- **Leading constant.** The chance that $27/(2\alpha) - \rho_p$ lies within its observed distance 1.79×10^{-5} of some fraction with denominator ≤ 6 is 4.3×10^{-4}, comparable to the muon offset.
- **Screening term.** The α^2 term is 63 times the model uncertainty, so some such term is required. Its form is not determined: the five fitting forms differ from the manuscript's by 1.3–2.6×10^{-7} in ρ_p, below the 2.8×10^{-7} carried by α. Separating the closest of them at 3σ would need $u(\alpha^{-1}) \approx 2 \times 10^{-9}$, ten times below the present value.

3.4 Dependence on the α input

α^{-1} input	Value	Model ρ_p	z, reference only	z, with α	Muon z
CODATA 2022, used in v13	137.035999177(21)	1836.1526734371	+0.35	+0.04	+0.39
Electron $g-2$, Fan et al. 2023	137.035999166(15)	1836.1526732886	−4.3	−0.67	+0.38
Rb recoil, Morel et al. 2020	137.035999206(11)	1836.1526738286	+12.6	+2.7	+0.40
Cs recoil, Parker et al. 2018	137.035999046(27)	1836.1526716686	−54.9	−4.8	+0.34
CODATA 2018	137.035999084(21)	1836.1526721816	−38.9	−4.4	+0.36

Reference m_p/m_e is CODATA 2022 in every row. The Cs and Rb determinations disagree with each other by about 5σ; the CODATA 2022 value combines them with the electron $g-2$ result and other inputs.

At their precision the lepton relations do not depend on which α is used. The proton relation does. It passes with the CODATA 2022 adjustment and with the electron $g-2$ value, and fails at −4.8σ with the Cs value and +2.7σ with the Rb value, counting the α uncertainty. With the CODATA 2018 α the same formula would have shown $z = -39$ in the units of Table 3.

3.5 Fitted and predicted

- **Forced by the data at the percent level:** the unit energy $\frac{3}{2}\alpha^{-1}$, the τ weight 17, the proton weight 9.
- **Selected at high precision among the enumerated alternatives:** the redistribution space 14 and $\eta = 1$ (muon, ppb level); the Compton subtraction, the term 1/3 and the α^2 term (proton).
- **Tested after selection:** the τ offset −17, to ±0.18, which excludes the neighbouring rule at 7σ; the proton α^2 coefficient, to ±1.6%.
- **Scale:** fixed to the infrared; Sec. 3.7 finds no other scale where the relations hold.
- **Not tested:** the form of the screening term at present α precision, and the use of α(0) with pole masses, since any other convention moves the results by more than the offsets.

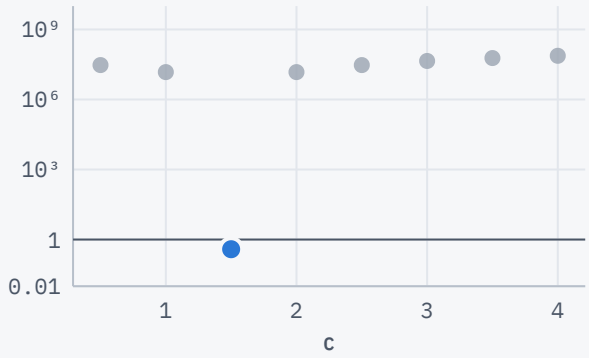
3.6 Simple-ratio neighbours of each choice

Figure 1 moves one choice at a time across its grid of simple ratios, with the others held at the manuscript's values, and plots the resulting pull. For the leptons the pull is the larger of the muon and τ pulls; for the proton it includes the uncertainty carried by α. Each manuscript value sits below 1σ. The nearest neighbour misses by at least $1.8 \times 10^4\sigma$ (redistribution dimension 15), and in four of the eight panels by more than $10^6\sigma$. On these grids there is nothing to tune: every discrete choice is fixed by the data, and only the split of the leading 3/2 between the Dirac level and the radius, $c/\lambda = 3/2$, is degenerate.

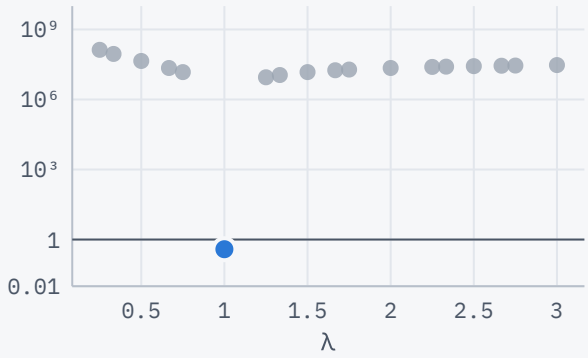
FIGURE 1 · PULL AT EACH SIMPLE-RATIO VALUE OF A CHOICE

Grid value Manuscript value

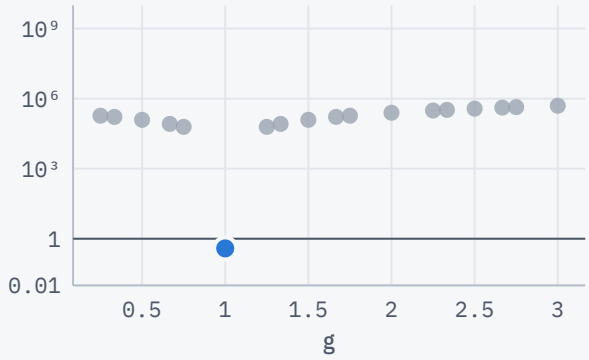
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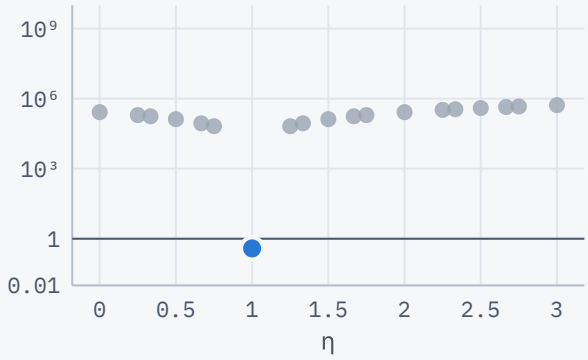
Radius factor λ · manuscript 1



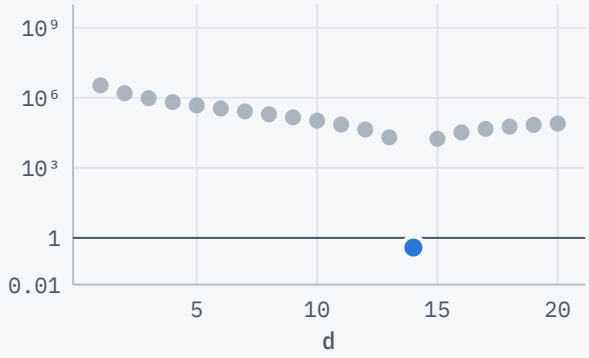
Structural weight g · manuscript 1



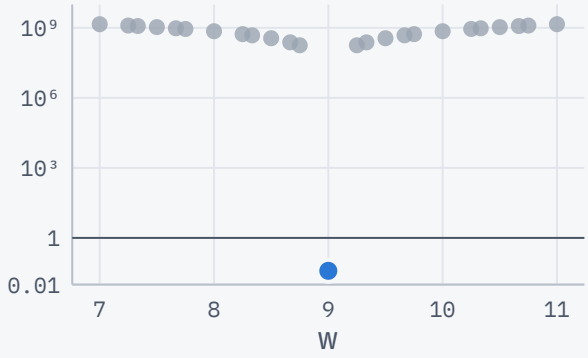
Calibration η · manuscript 1



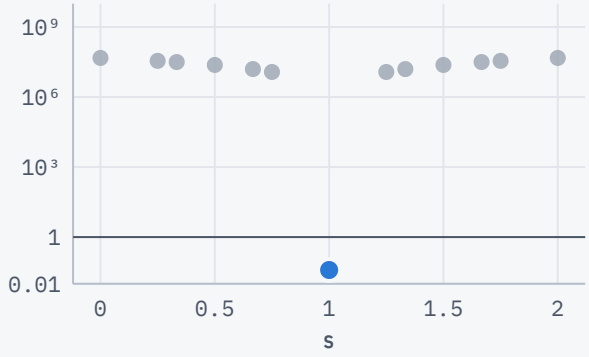
Redistribution dimension d · manuscript 14



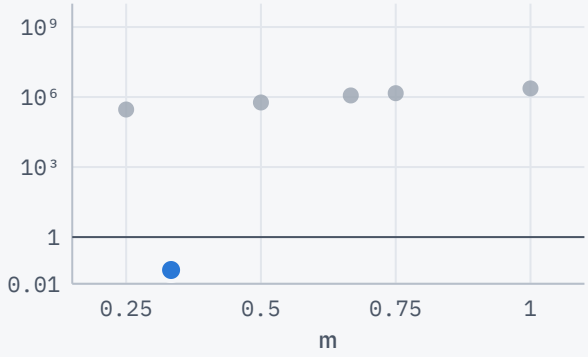
Colour weight W · manuscript 9



Reference length s · manuscript 1



Mean-charge term m · manuscript 1/3

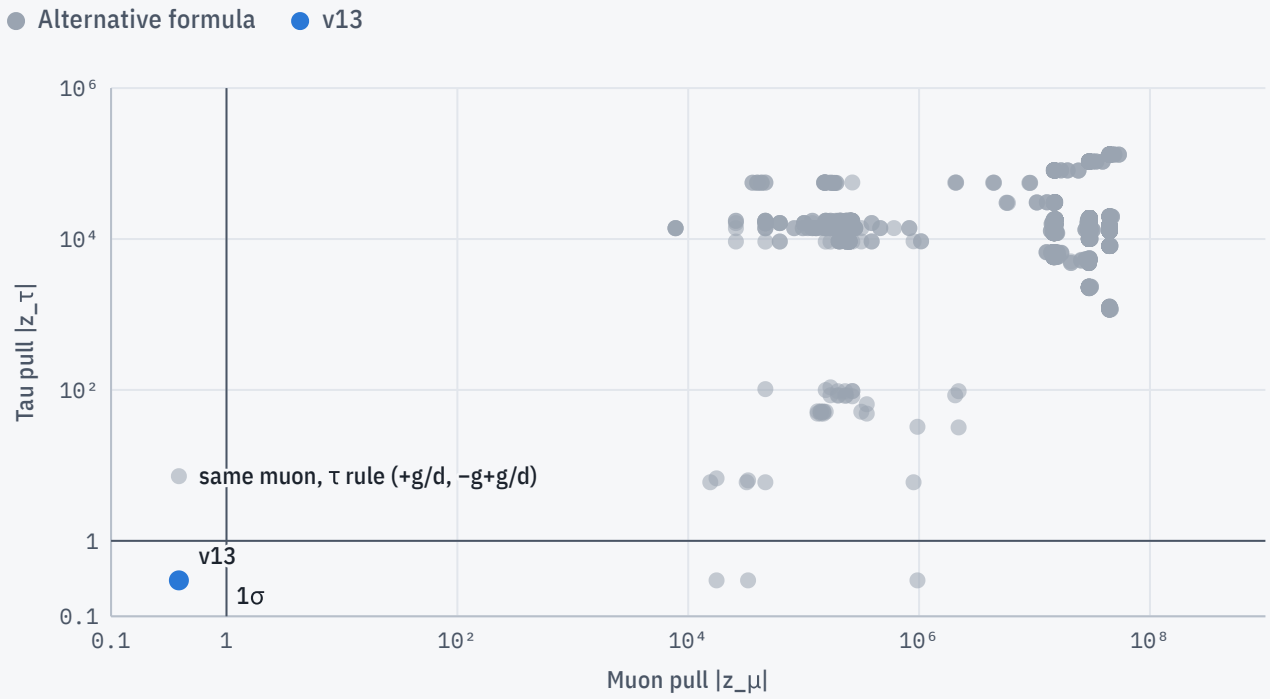


Vertical axis logarithmic; the horizontal hairline marks 1σ. Grids: fractions with denominator ≤ 4 (≤ 2 for the Dirac level c), integers for the dimension d. Lepton panels vary one of c, λ, g, η, d; proton panels one of W, s, m.

Choice	Manuscript	Pull	Nearest grid values: pull
Dirac level c	3/2	0.385	1: 1.49×10^{7}; 2: 1.49×10^{7}
Radius factor λ	1	0.385	3/4: 1.49×10^{7}; 5/4: 8.94×10^{6}
Structural weight g	1	0.385	3/4: 6.21×10^{4}; 5/4: 6.21×10^{4}
Calibration η	1	0.385	3/4: 6.6×10^{4}; 5/4: 6.6×10^{4}
Redistribution dimension d	14	0.385	13: 2.03×10^{4}; 15: 1.76×10^{4}
Colour weight W	9	0.0389	35/4: 1.79×10^{8}; 37/4: 1.79×10^{8}
Reference length s	1	0.0389	3/4: 1.18×10^{7}; 5/4: 1.18×10^{7}
Mean-charge term m	1/3	0.0389	1/4: 2.92×10^{5}; 1/2: 5.84×10^{5}

Figure 2 places each of the 3240 discrete formulas of Sec. 3.2 by its pulls from the muon and τ data. One sits inside the 1σ box. The next shares the manuscript's muon offset, changes the τ rule, and sits at 7.2σ in the τ. Every other formula is beyond $10^{4}\sigma$ in the muon or the τ. Figure 3 counts the formulas of both families within a given pull.

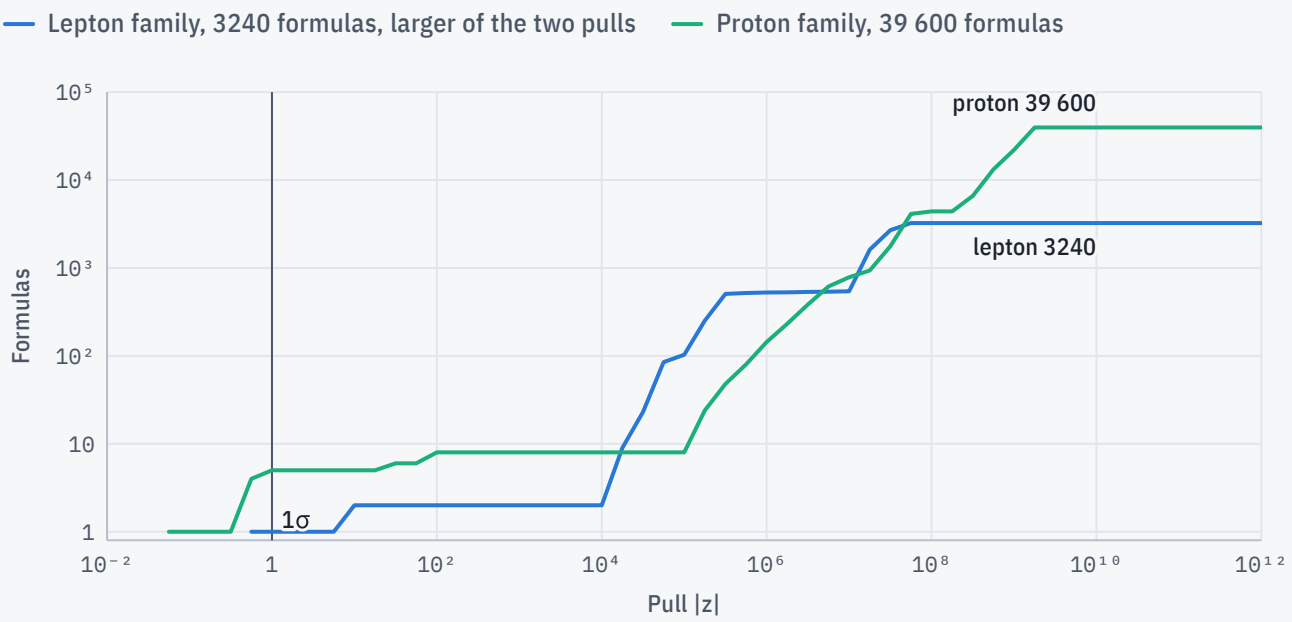
FIGURE 2 · RESULT SPACE OF THE LEPTON FORMULA FAMILY



Each dot is one distinct output pair; 1638 distinct pairs from 3240 formulas. Pulls below 0.1 are drawn at 0.1. Hover a dot for the formula.

Formula	|z_μ|	|z_τ|
c=3/2, dim S=4, g_τ=17, d=14, η=1, rule (+g/d, −g)	0.39	0.30
c=3/2, dim S=4, g_τ=17, d=14, η=1, rule (+g/d, −g+g/d)	0.39	7.19
All others: larger pull above	1×10⁴	

FIGURE 3 · FORMULAS WITHIN A GIVEN PULL



Cumulative count of formulas whose pull is below the value on the horizontal axis. Proton pulls include the uncertainty carried by α.

Pull below	Lepton formulas	Proton formulas
1	1	5
10	2	5
100	2	8
1×10^{4}	2	8
1×10^{6}	526	144
1×10^{8}	3240	4400
1×10^{10}	3240	39600

3.7 Running: is there a better scale than the infrared?

The manuscript uses α at $q^2 = 0$ and pole masses. To test whether another energy suits the relations better, each relation was written as the offset it needs at a renormalization scale μ,

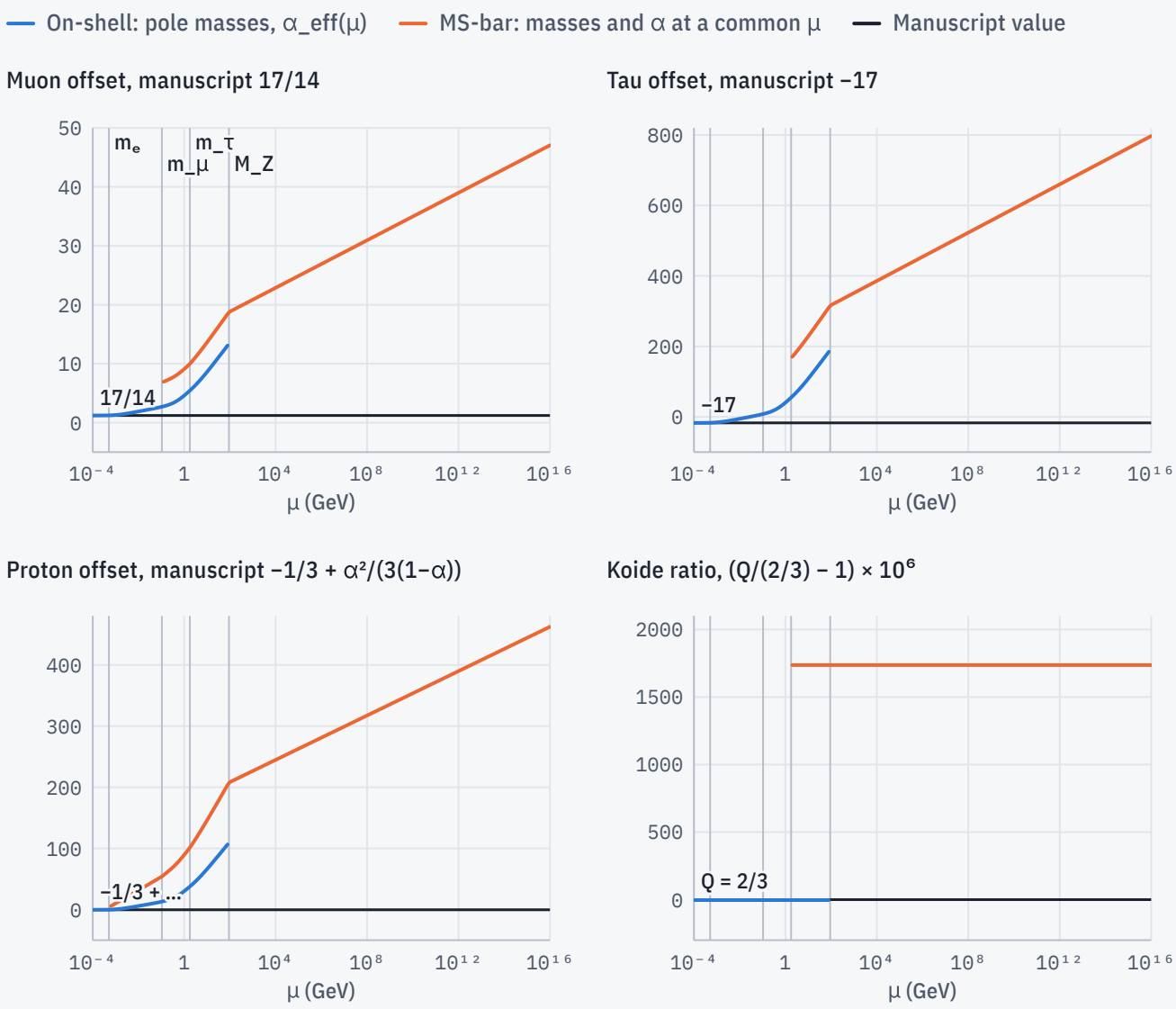
$$o_\mu = \frac{m_\mu}{m_e} - \frac{3}{2\alpha(\mu)}, \qquad o_\tau = \frac{m_\tau}{m_e} - \frac{51}{2\alpha(\mu)}, \qquad o_p = \frac{m_p}{m_e} - \frac{27}{2\alpha(\mu)} + \frac{27}{2},$$

and evaluated in two schemes.

- **On-shell.** Pole masses and the effective coupling $\alpha_{\text{eff}}(\mu) = \alpha/(1 - \Delta\alpha(-\mu^2))$: exact one-loop leptonic vacuum polarization, and a dispersive hadronic part built from narrow ρ, ω, φ, J/ψ, ψ(2S) and Υ resonances with their measured Γ_{ee} plus perturbative continua, calibrated to $\Delta\alpha^{(5)}_{\text{had}}(M_Z^2) = 0.02783$ (PDG 2024). The band shows the spread for 0.02761 to 0.02795 and for a light-quark continuum R from 2.0 to 2.3.
- **$\overline{\text{MS}}$.** Running masses at a common scale and $\hat{\alpha}(\mu)$: leptonic running at two loops with decoupling, the hadronic part rescaled to $\hat{\alpha}^{(5)}(M_Z)^{-1} = 127.930$, and above M_Z one-loop Standard Model running of

α_{em} with the lepton mass ratios held fixed, since the gauge terms in their running are flavour-universal.

FIGURE 4 · OFFSETS EACH RELATION NEEDS, AGAINST THE SCALE M



Scale μ in GeV, logarithmic; vertical hairlines at m_e, m_μ, m_τ and M_Z. On-shell curves stop at M_Z. The $\overline{\text{MS}}$ curves start where all the leptons involved are active. In the Koide panel the on-shell value is the measured one and the line at 0 is Q = 2/3.

Scale	On-shell o_μ	o_τ	o_p	MS-bar o_μ	o_τ	o_p	MS-bar Koide (ppm)
μ → 0	1.214	-17.05	-0.3333	–	–	–	–
m_e	1.243	-16.56	-0.07363	–	–	3.932	–
m_μ	2.694	8.103	12.98	6.827	–	54.1	–
m_τ	5.441	54.79	37.7	9.98	166.7	101.3	1737
10 GeV	8.605	108.6	66.19	13.63	228.7	145.9	1737
M_Z	13.39	190	109.3	18.76	316	207.7	1737
10^{16} GeV	–	–	–	47.06	797	462.4	1737

- **On-shell.** The offsets equal the manuscript's values only as μ → 0 and grow monotonically with the scale: muon 1.214, 2.69, 5.44, 13.4; τ −17.05, +8.1, +54.8, +190; proton −0.333, +13.0, +37.7, +109, at μ → 0, m_μ, m_τ and M_Z. The relations stay within 1σ only below μ ≈ 6 keV (muon), 300 keV (τ) and 0.5 keV (proton), all below the electron mass.
- **Offset rule.** The manuscript ties the two lepton offsets by $o_\tau/o_\mu = -d = -14$. The ratio is −14.01 as μ → 0, −11.9 at 1 MeV, −2.9 at 10 MeV and +2.9 at 100 MeV.
- **$\overline{\text{MS}}$.** The QED mass anomalous dimension is the same for e, μ and τ, so ratios at a common scale stop running above the τ: $\bar{m}_\mu/\bar{m}_e = 210.66$, $\bar{m}_\tau/\bar{m}_e = 3578.2$. With $\hat{\alpha}(\mu)$ still running, the muon offset goes from 6.8 at m_μ through 10.0 and 18.8 at m_τ and M_Z to 47 at 10^{16} GeV, and the τ offset from 167 to 797. Neither approaches 17/14 or −17: the pole-to-$\overline{\text{MS}}$ conversion alone raises m_μ/m_e by 1.9%.
- **Reading.** No scale suits the relations better than the infrared. A continuous scale can place one offset on any chosen value (the on-shell muon offset passes 3/2 at 2.6 MeV, 2 at 14 MeV and 3 at 200 MeV), which is the flexibility counted earlier in this section. Two relations take simple values at a common scale only as μ → 0, with α at its Thomson value and pole masses. Koide behaves the same way: with $\overline{\text{MS}}$ masses it is off by $+1.74 \times 10^{-3}$ at every scale above m_τ, consistent with Xing and Zhang.

4 Koide's ratio across the parameter space

With $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$, the manuscript evaluates $Q/(2/3) - 1$ as a diagnostic (Eq. 8.3). Its range is $1/3 \le Q \le 1$; $Q = 1$ would need two vanishing masses, so the question of interest is where $Q/(2/3) = 1$.

4.1 At the manuscript's point

The relations give $Q/(2/3) - 1 = -1.02 \times 10^{-6}$; the measured masses give $(-3.3 \pm 7.6) \times 10^{-6}$, the uncertainty set by the τ mass. The model's value comes from its fit to the masses. Its $m_\tau = 1776.957$ MeV and the τ mass that makes Koide exact with the measured e and μ, 1776.969 MeV, differ by 0.012 MeV, 0.13σ of the PDG mass. A τ mass measured to about 0.004 MeV would separate the two at 3σ.

4.2 What can move Q

With the choices kept on simple ratios, Koide's ratio can move only through the τ weight in steps of the grid, through α, and through the scale.

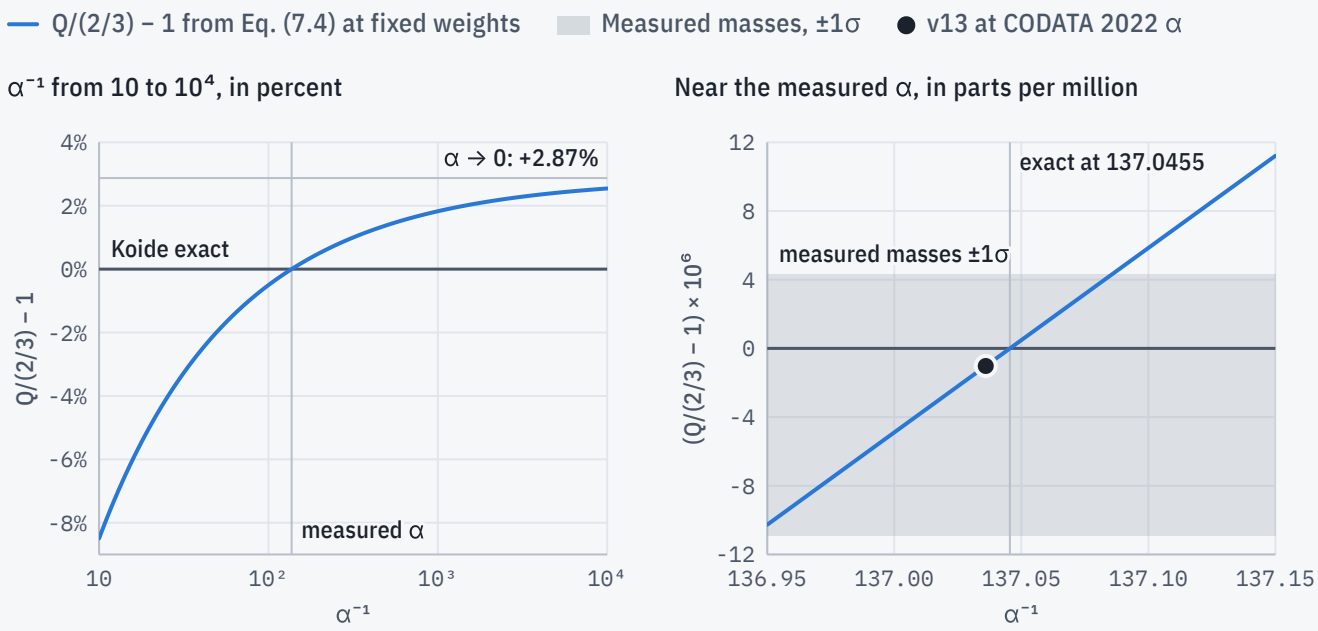
Change	Q/(2/3) − 1	Larger lepton pull
τ weight 9 ($g = 1/2$)	−9.5%	1.2×10⁵σ
τ weight 13 ($g = 3/4$)	−4.0%	6.2×10⁴σ
τ weight 17 ($g = 1$), manuscript	−1.0×10⁻⁶	0.39σ
τ weight 21 ($g = 5/4$)	+3.1%	6.2×10⁴σ
τ weight 25 ($g = 3/2$)	+5.7%	1.2×10⁵σ
α, continuous	0 at α⁻¹ = 137.0455	3112σ; α moved by 4.5×10⁵ u(α⁻¹)
Scale, on-shell	unchanged	pole masses do not run
Scale, $\overline{\text{MS}}$, μ ≥ m_τ	+1.74×10⁻³	same at every μ

No admissible change makes Koide exact. The grid neighbours of 17 move Q by percents, α would have to move by 4.5×10^5 times its uncertainty, and the scale moves Q away from 2/3. Exact Koide would need a τ weight of 17.000115, which is not a simple ratio: the weight is $1 + (\mathrm{tr}\,\Sigma^2)^2$, an integer.

4.3 Koide's ratio as a function of α

At fixed weights the ratio drifts with α and crosses 2/3 once, at $\alpha^{-1} = 137.0455$. As α → 0 it tends to $18/(1+\sqrt{17})^2 = 0.6858$, 2.87% above 2/3. In this model Koide's value therefore depends on the electron term at finite α. With every mass linear in $1/\alpha$ and the electron as the unit, $Q = 2/3$ cannot hold for all α: $\mathrm{Tr}\,M - \frac{2}{3}(\mathrm{Tr}\sqrt{M})^2$ contains the term $-\frac{4}{3}(1+\sqrt{G})\sqrt{\kappa}$, G the τ weight, which no choice of constant offsets removes.

FIGURE 5 · KOIDE'S RATIO AGAINST A⁻¹, MANUSCRIPT FORMULAS

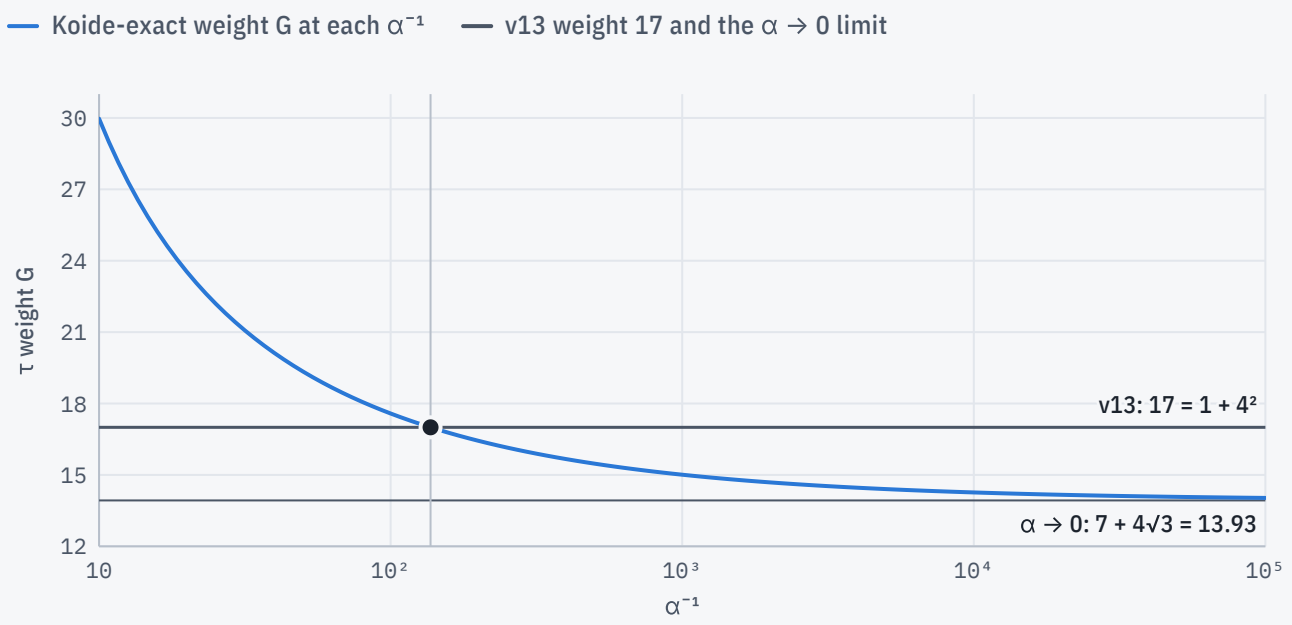


α⁻¹	Q/(2/3) – 1
10	-8.496%
50	-1.962%
100	-0.505%
137.036 (CODATA 2022)	-1.02 ppm
137.0455	-0.00 ppm
300	0.947%
1000	1.825%
10000	2.543%
α → 0	+2.872%
Measured masses	-3.3 ± 7.6 ppm

Figure 6 gives the τ weight G that would make Koide exact at each α, keeping the manuscript's offset rule $(+G/14, -G)$. It runs from 17.58 at $\alpha^{-1} = 100$ through 17.0001 at the measured α to $7 + 4\sqrt{3} = 13.93$

as α → 0, the value for which $\sqrt{m_\tau/m_\mu} = 2+\sqrt{3}$. The limit is irrational, so no integer weight $1+n^2$ reaches it. The weight 17 meets the curve only at $\alpha^{-1} = 137.0455$.

FIGURE 6 · T WEIGHT THAT MAKES KOIDE EXACT



α^{-1}	Koide-exact τ weight G
10	29.9692
31.6	21.0725
100	17.5803
316	15.8929
1000	15.0087
3160	14.5288
1×10⁴	14.2638
3.16×10⁴	14.1163
1×10⁵	14.0338
137.036 (CODATA 2022)	17.000116
α → 0	13.9282

4.4 A geometry with Koide built in

Koide's ratio is a statement about $\sqrt{M}$. $Q = 2/3$ holds exactly when the vector $(\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ makes 45° with $(1,1,1)$ (Foot 1994), that is, when the traceless part of $\sqrt{M}$ has the same Hilbert–Schmidt norm as its scalar part. A Z_3-circulant square root,

$$\sqrt{M} = \mu\left[I + \tfrac{1}{\sqrt{2}}\left(e^{i\delta}T + e^{-i\delta}T^\dagger\right)\right], \qquad T \text{ the cyclic shift of the three profiles,}$$

has eigenvalues $\mu\left(1+\sqrt{2}\cos(\delta + 2\pi k/3)\right)$ and gives $Q = 2/3$ for every μ and δ; the masses then fix δ. This parametrization is standard in the Koide literature. The manuscript's carrier has neither ingredient. Its balance between scalar and nonscalar parts is imposed on M, through the graph norm on $\mathcal{W}$, and its

profiles sit in blocks of dimensions 1, 14, 1 with no symmetry that exchanges them. A carrier with Koide exact would need three sectors permuted by a cyclic symmetry and the 45° condition on the square-root operator. The angle is 44.99997° for the manuscript's masses and 44.99991(22)° for the measured ones.

5 Findings of the mathematical check

MODERATE The proton comparison omits the uncertainty carried by α

Sec. 10.5, Table 3, abstract, Sec. 11.3

Since $\partial\rho_p/\partial\alpha^{-1} = 27/2$, the uncertainty $u(\alpha^{-1}) = 2.1 \times 10^{-8}$ gives $u(\rho_p) = 2.8 \times 10^{-7}$, which is 8.9 times the CODATA $u(m_p/m_e) = 3.2 \times 10^{-8}$ used for z. Against the model uncertainty the residual 1.11×10^{-8} is 0.04σ. Equivalently, Eq. (10.12) solved for α gives $\alpha^{-1} = 137.0359991762(24)$, consistent with CODATA. The statement in Sec. 8.2 that input uncertainties are negligible holds for the leptons only. The sentence "at the central input values" is accurate; the test is limited by α. A model-uncertainty column in Table 3, or the implied value of α^{-1}, would state this. Sec. 3.4 shows how the comparison changes with the α input.

MINOR The m_pc^2 row is not an independent test

Table 3

The model value uses the same m_ec^2 as the CODATA value, whose uncertainty of 0.29 eV equals the m_e contribution $\rho_p\, u(m_ec^2) = 2.9 \times 10^{-7}$ MeV. The quoted $z = 0.0103$ is at the level of the rounding of the reference: the unrounded product $\rho_p^{\mathrm{CODATA}}\, m_ec^2 = 938.2720894273$ MeV gives $z = 0.020$.

MINOR The return map is fixed only on active inputs

Sec. 6.2, Eqs. (6.5)–(6.7)

The trace condition is imposed for positive active inputs. The map $\mathscr{J}'(Z) = \mathscr{J}(Z) + c\,\langle s, Zs\rangle P_\mu/14$ with $c \geq 0$ is completely positive, G-covariant, maps into $P_\mu\mathcal{W}$ and satisfies $\mathrm{Tr}\,\mathscr{J}'(Z) = \mathrm{Tr}(\mathcal{D}Z)$ on active inputs. The claim "these conditions determine the return map" needs $\mathscr{J}(P_e) = 0$ or the trace condition on all inputs; the latter also gives $\mathrm{Tr}\,\mathscr{L}(Z) = 0$ in Eq. (6.7) for every Z. No result changes, since $Z_{\mathfrak{u}}(0) = A$ has no component on the scalar line.

COSMETIC τ input sensitivity is 8.30×10⁻⁷ MeV

Sec. 8.2

$$\begin{aligned} &25.5\, m_ec^2\, u(\alpha^{-1}) + \rho_\tau\, u(m_ec^2) \\ &= 2.736 \times 10^{-7} + 5.564 \times 10^{-7} = 8.30 \times 10^{-7}\ \mathrm{MeV}. \end{aligned}$$

The printed "at most 8.31×10⁻⁷" remains a valid bound.

COSMETIC

Reused symbols

Secs. 5, 6, 10, App. B

A is both the response operator and the Clifford index in Eqs. (5.2)–(5.7), with both meanings inside Eq. (5.6). B is an index in Eq. (5.2) and the correction operator. u denotes a spin lift, the phases u_e, u_τ, the survival fraction $u(\zeta)$ and the flow parameter of Eq. (10.11). t is the trace 31 and the generators t^a; r is the radius and the cardinality in Eq. (B.2). Only Eq. (5.6) is ambiguous as printed.

6 Verification record

Item	Equations	Method	Result
Shape functional	(2.1)	Koszul formula for diagonal left-invariant metrics; energy density $\frac{1}{4}\sum_i (a_j + a_k - a_i)^2$	Agrees
Curvature bound, Killing spinors	(2.2)–(2.6)	Lichnerowicz identity and Friedrich bound	Agrees
Clifford carrier, Hodge splitting	(3.1)–(3.6)	Numerical, matrices (B.1)	Agrees
Stabilizer, profiles, commutants	(4.1)–(4.4), (A.3)	Numerical; Spin(4) commutant dimension	10, agrees
Response operator	(5.1)–(5.9)	Numerical	Spectrum {0, 1, 17}, multiplicities 1, 14, 1
Minimum	(6.1)–(6.4)	Analytic; reduces to minimizing $16z_\tau$ with $14z_\mu + z_\tau = 31$, $z \geq 0$	$Z_* = \frac{31}{14} P_\mu$
GKSL semigroup, Kraus form	(6.5)–(6.8), (9.4)	Matrix exponential of the 256×256 generator against the Kraus map, random inputs	Agrees
Normalized and variational flows	(6.9)–(6.10), (9.9)–(9.11)	Finite differences	Agrees
Dilation and covariance	(9.5)	Random $U \in G$	Agrees
Collective profiles, collision step	(9.6)–(9.8)	Numerical	Agrees
Field equations, decay width	(9.2)–(9.3), (9.14), (9.19)–(9.20)	Sign conventions, standard forms	Agrees
Colour sector	(10.1)–(10.10), (10.13)–(10.14)	Numerical on $(\mathbb{C}^3)^{\otimes 3}$	Agrees
Casimirs	(A.1)–(A.2)	Numerical	Agrees
Numerics	Tables 1, 3; (8.3); η interval; thresholds; (11.4)	40-digit arithmetic	Agrees, except finding 4

7 References for the context sections

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Nambu 1952, Mac Gregor 2005, Kritov 2021, Scott 2023 and Foot 1994 are cited as in the manuscript's reference list, items [5], [6], [7], [26] and [14].

Prepared with Claude Code from the v13 PDF. Numerical checks used the explicit matrices of Eq. (B.1) and the inputs of Eq. (8.2). Scale and scheme figures in Sec. 2.2 are one-loop, leading-order estimates. Figures 1–6 are computed from the manuscript's formulas with the same inputs.